

Introduction

- Oral diseases affect **~3.7 billion people worldwide** [1]
- Untreated conditions can reduce **quality of life**
- Many cases are **preventable**, but care is delayed due to [2][3]
 - Cost
 - Limited access
 - Lack of awareness
 - Dental anxiety
- Early detection is critical** Enables simpler treatment
 - Improves long-term outcomes
- Traditional diagnosis requires **clinic visits & equipment**
- Proposed solution:**
 - Mobile-based screening system
 - Image-based deep learning + risk questionnaire
 - Improves **accessibility and early detection**

Objective

- Develop a **hybrid system** for detecting common oral conditions
- Targets: calculus, caries, gingivitis, tooth discoloration
- Combines:
 - Image-based deep learning
 - Questionnaire-based risk assessment
- Designed to be **accessible and low-cost**
- Supports **early detection and awareness** [4]
- Focuses on people with **limited access to dental care**

System Design

- Client Server Architecture:**
 - User inputs: oral image + questionnaire
- Image** → deep learning model
- Responses** → risk assessment decision tree
- Outputs combined via **fusion layer**
- Risk results + personalized recommendations shown to user

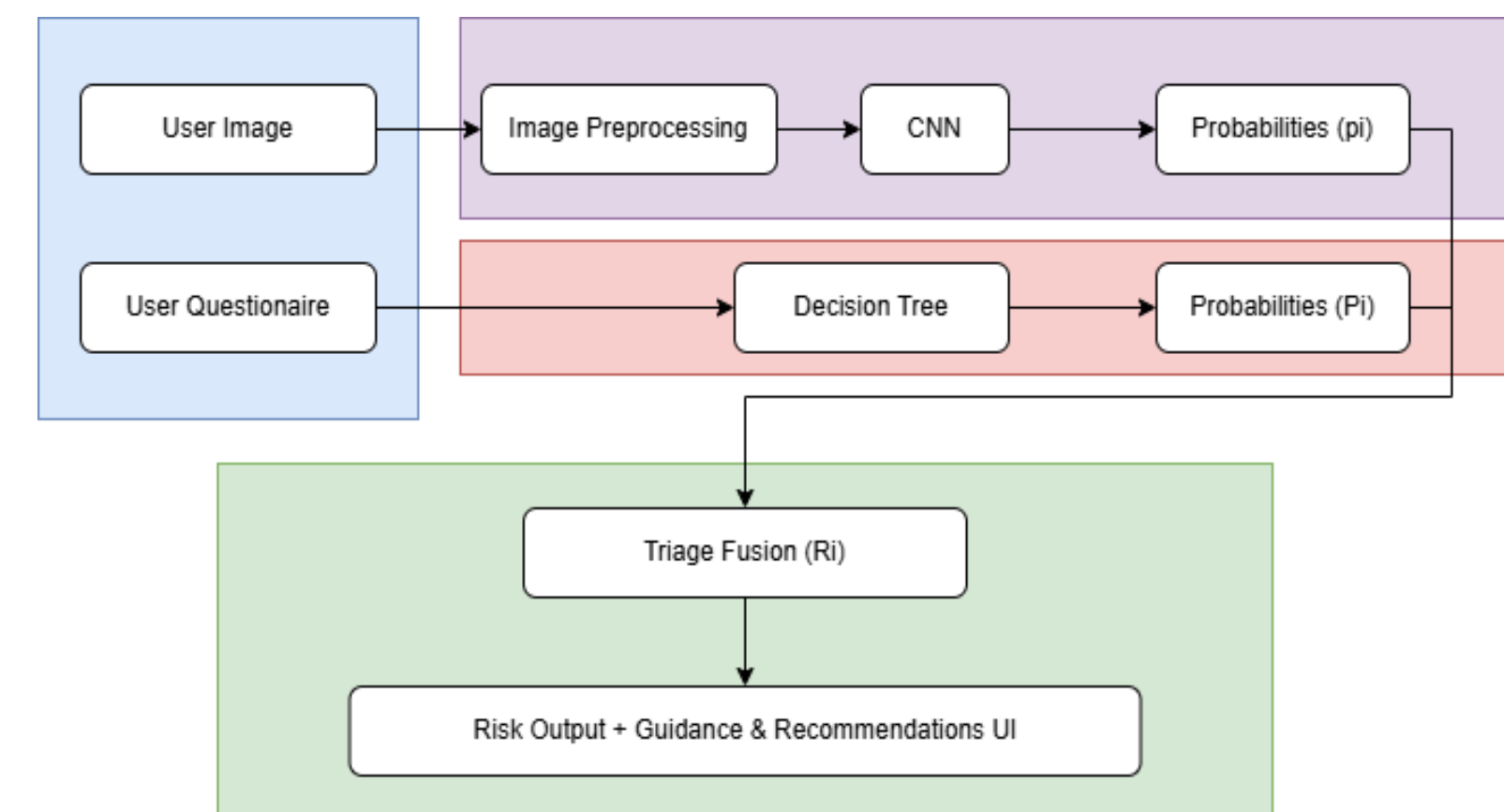


Figure 1. System Architecture highlighting components and workflow

Preprocessing

- Input validation stage** ensures reliable predictions
 - Image quality checked (clarity, contrast)
- Pre-filter verifies **teeth and gumline presence**
 - Invalid images are **rejected early**
 - Valid images sent to **classification model**
- Reduces false predictions, improves reliability

Visual Model (Deep Learning)

- Visual model based on **MobileNetV3 Small (transfer learning)**
- Designed for efficient oral image analysis
- Images processed:
 - Resized to **224 × 224**
 - Normalized (ImageNet statistics)
 - Augmented (rotation, flipping, brightness)
- Architecture:
 - Frozen backbone**
 - Custom classifier head
- Uses **masked multi-label approach**
 - Relevant labels used; Unrelated labels ignored (-1)
 - Trained with **masked binary cross-entropy**
- Optimized using Adam optimizer for stable learning

Questionnaire

- Questionnaire model estimates **risk from patient inputs**
- Inputs: hygiene, sugar intake, dental visits, symptoms, history etc.
- Responses form a **weighted risk score**
 - Converted to probabilities using **sigmoid function**
- Captures **non-visual risk factors** to increase generalizability
- Combined with image model via **weighted fusion**
- Outputs **final risk predictions**

Results

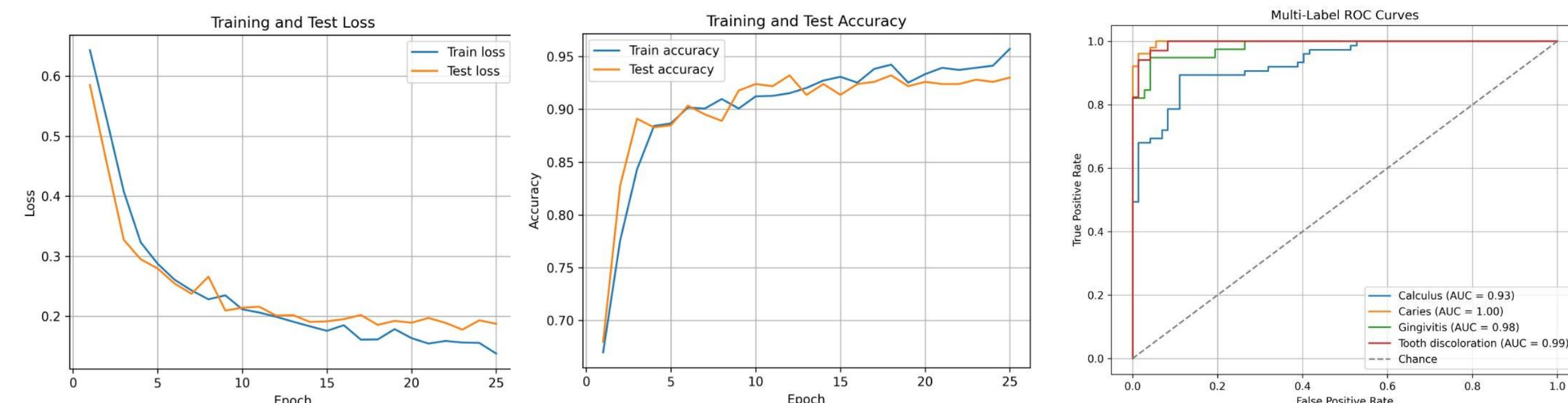


Figure 2. Evaluation Metrics for Visual Model. (a) Training and Test Loss, (b) Training and Test Accuracy, (c) Multi-label ROC Curves. Average 92% Model Accuracy and 0.97 Average AUC across the four conditions.

Frontend Integration

- Mobile-First Design
 - Intuitive, accessible experience using **React Native (Expo Go)**
- Frontend (React JS/TS)** handles UI, collects images and questionnaire responses
 - Sends data to backend via secure **REST API**
- Backend (FastAPI)** performs image preprocessing and AI inference efficiently
 - Generates probability scores and detected oral conditions before sending back to the front end for display
- Results Display shows risk levels, detected conditions, and recommendations
- Testing & Iteration:** paper prototypes → Figma → working application
 - User Feedback collected continuously to improve interface and workflow
- Scalability, Maintainability & Focus**, separation simplifies updates, ensures user-centric design

UI / UX

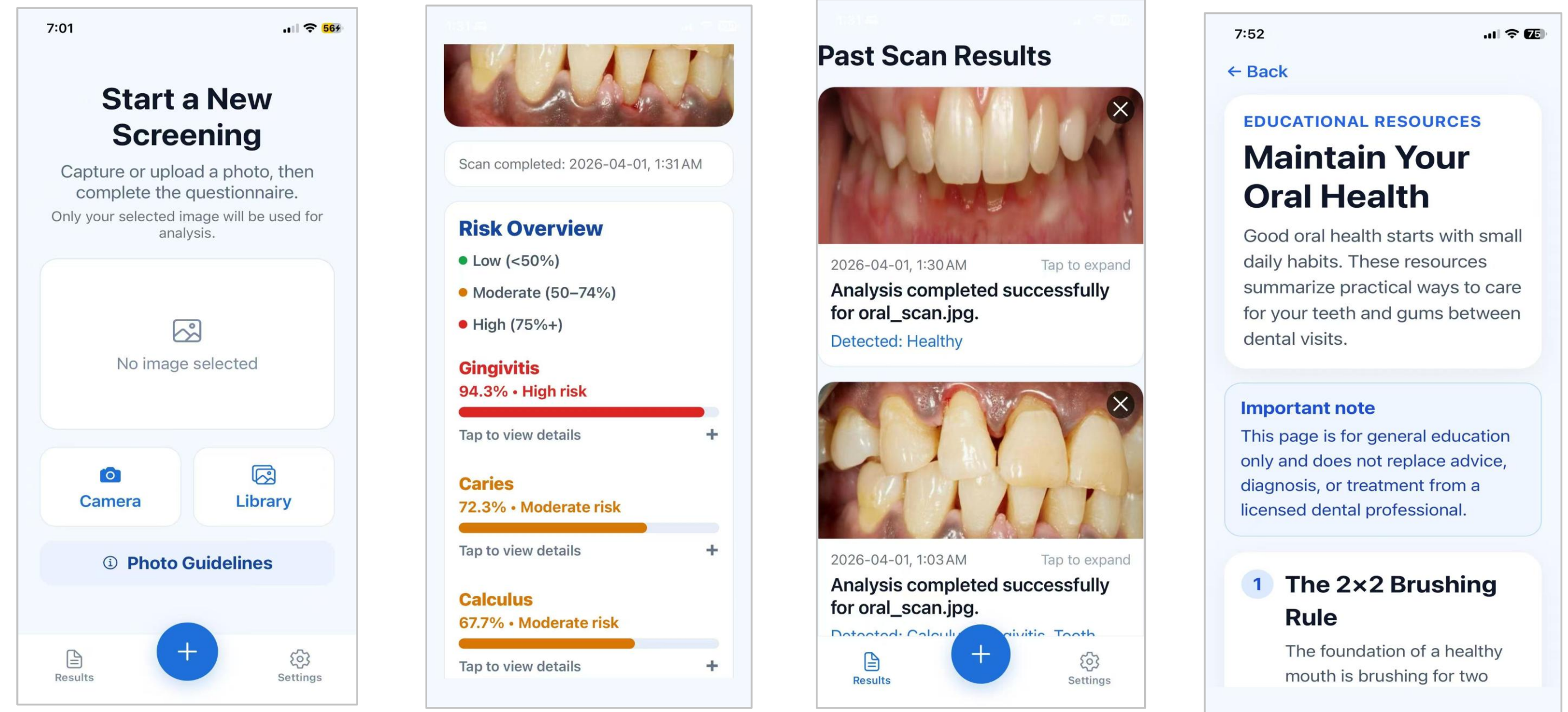


Figure 3. SELFIE Application User Interface. (a) Landing page, (b) Results display, (c) Patient history tracking, (d) Educational resources. The interface is designed to provide accessible screening, result interpretation, and user guidance.

Future Considerations

- Future work will focus on **expanding the dataset** with fully labeled conditions, improving model performance and learning relationships between co-occurring diseases.
 - Addition of more oral conditions, extending diagnostic capabilities.
- Incorporating **condition-specific thresholds** to reduce false positives while maintaining high sensitivity.
- Future development may include **real-world deployment**, user interface design improvement, and system integration into mobile health platforms for broader accessibility.